



Effects of game-based learning supports on students' math performance and perceived game flow

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Abstract

Adopting a pretest–posttest experimental design with repeated measures, this study examined the effects of three types of game-based learning supports in the form of modeling on knowledge development that contributed to successful math problem solving and students' perceived game flow. Forty-one sixth-grade students participated in the study and played a 3D architecture game that aims to promote mathematical conceptual understanding and problem solving skills, and presented with different types of learning supports (i.e., conceptual knowledge only, procedural knowledge only, and the combination of the two). No significant effect of the types of supports was found on participants' perceived game flow. However, there was a significant impact of support type on participants' performance on two posttests. The results indicated that the learning support with procedural knowledge only, compared with the learning support with both conceptual and procedural knowledge as well as that with conceptual knowledge only, was significantly more effective in promoting students' knowledge acquisition related to mathematical problem solving.

Keywords Game-based learning · Learning support · Modeling · Math performance · Game flow

Introduction

Video games are considered a promising approach for learning in K-12 settings, but the effectiveness of educational video games in comparison with conventional instructional approaches on learning outcomes are mixed (Bainbridge et al., 2022; Habgood & Ainsworth, 2011; Hartt, et al., 2020). One of the central challenges of educational video games is helping learners transform intuitive learning into generative learning during game play (Fiorella & Mayer, 2012; ter Vrugte et al., 2015). A possible solution to this challenge is adding learning support (Lee & Ke, 2019; O'Neil et al., 2014; Wouters et al., 2009) that

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refers to assistance, guidance, or instruction that stimulates or facilitates students' game-based learning (GBL) processes (Tobias, 2009; Tobias et al., 2011). For example, in-game references, as well as cues and hints are the two frequently used approaches to delivering the learning support in GBL (Pan et al., 2022). In-game references could be embedded as a game object situated in the game world, such as a "guidebook" or a "journal" that summarizes game dialogue and math concepts. Cues and hints are presented through the head-up display, task-oriented display, or an end-of-level performance summary (Ke, 2016; Pan et al., 2022).

An increasing number of studies indicate that learning embedded in educational video games is the crucial component to determine the educational effectiveness of games rather than the innate characteristics of digital games (e.g., Bainbridge et al., 2022; Greitemeyer & Mügge, 2014; Leemkuil et al., 2003), such as narrative, aesthetics and system mechanics. These researchers argued that simply using games for learning imposes considerable constraints because it is "difficult to integrate deep content, strategies, and skills" (Graesser et al., 2009, p.12) in the game. Therefore, they apply a *value-added* approach to GBL (Clark et al., 2016), aiming to investigate the effects on learning quality of adding features to an educational game. Providing learning support on GBL is a type of value-added approach adopted to examine the potential benefits of learning quality of adding features. Empirical studies revealed that it helped students transform implicit knowledge to explicit knowledge (ter Vrugte & de Jong, 2017), transfer to assessments outside of the game (Bainbridge et al., 2022; Barzilai & Blau, 2014), transform intuitive processing to generative processing (O'Neil et al., 2014), and externalize problem representation (Lee & Ke, 2019).

However, prior research suggests that adding learning support to games is complex (Bainbridge et al., 2022; ter Vrugte et al., 2015; Wouters & van Oostendorp, 2013). It was not always effective in terms of knowledge acquisition (Moreno & Mayer, 2005; ter Vrugte et al., 2015) and it was reported to inhibit the game flow (Erhel & Jamet, 2013; ter Vrugte et al., 2015; Vandercruysse et al., 2016). More work needs to be done to better understand how to successfully design learning support that promotes learning while maintaining students' game flow (Shute et al., 2019). Therefore, the current study aims to investigate the design and impact of variant game-based learning supports on students' math achievement and game flow. Specifically, considering the paramount roles of conceptual (i.e., semantic structure and mathematical relationships) and procedural knowledge (i.e., the computational skills) on mathematical problem solving, in the current study, we investigate how the learning support of conceptual and procedural knowledge for mathematical problem solving in the form of modeling in GBL affects students' performance and game flow.

Learning support in game-based learning

From the cognitive load perspective, the purpose of integrating learning support in games is to help students overcome the limitations of human cognitive architecture (i.e., short-term memory, Mayer, 1985, 2014; Paas et al., 2003; Sweller, 1988, 2005). GBL environments are complex and challenging that demand cognitive efforts that heavily rely on the working memory (Wouters & van Oostendorp, 2013). Students can get stuck and feel frustrated when playing the game. Providing learning support can reduce the cognitive load during information processing in a complex gaming environment, and consequently foster learning (Wouters et al., 2009).

However, the literature on the design of effective learning support that will always foster learning in general is inconclusive. On the one hand, some researchers reported that extra learning support did not produce learning improvement (Clark et al., 2016). On the other hand, a recent Bayesian meta-analysis found that learning support significantly impacted learning in STEM education (Kim et al., 2018). This finding is in line with Wouters and van Oostendorp's (2013) meta-analytic review, where they found a positive effect size ($d=0.34$, $p<.001$) of learning support in GBL. Researchers argued that such controversial results were attributed to the fact that many factors, such as characteristics of the learning support (e.g., Vandercruysse et al., 2016), characteristics of the students (e.g., ter Vrugte et al., 2015), and characteristics of game genres (e.g., ter Vrugte & de Jong, 2017), affect the effectiveness of game-based learning support. Therefore, rigorous research is needed to provide guidance on how to design effective learning support in GBL.

Modeling as learning support in game-based learning

Modeling is one of the broadly designed learning supports for educational video games in recent years (Ke, 2016; Shute et al., 2019; Wouters & van Oostendorp, 2013). It is an instructional strategy used to facilitate the problem-solving processes by providing explications or examples, consisting of a problem formulation, the procedural steps, and the final solution to demonstrate the way to solve the problem (Renkl, 2002; Shute et al., 2019). Scaffolding and worked-out examples are two most common forms of modeling (Lee & Ke, 2019). A number of studies provided evidence on the effectiveness of modeling in classroom instruction (e.g., Bentley & Yates, 2017; Cooper & Sweller, 1987). Learners, especially novices, could develop a mental representation of the problem by observing an expert performing a complex task (e.g., Collins et al., 1989; Wouters et al., 2009). Modeling is reported to be most effective in well-structured domains such as mathematics and programming (Bentley & Yates, 2017; Renkl & Atkinson, 2003). Nevertheless, for modeling employed in GBL, the empirical evidence is limited, inconsistent, and inconclusive (Mayer, 2011; Wouters, 2017).

First, a criticism on using modeling in GBL is that students might not be actively engaged in studying the examples (Renkl et al., 1998; van Merriënboer & Kirschner, 2018). For example, Vandercruysse et al. (2016) investigated the effects of learning supports with different delivery methods on participants' math performance and intrinsic motivation in GBL. The learning support was in the form of modeling, which provided the relevant problem solutions that included a problem statement, action sequences, and a final solution. One hundred twenty-two 15- to 19-year-old prevocational students were randomly assigned to a math game with modeling that was embedded in the game world (ICC), the same game with learning support that was delivered in stand-alone paper handouts (ECC), and the game without support (control condition). This study reported that students in the condition of ECC and the control condition showed similar performance. The authors reported that students in ECC condition did not use the learning support. This finding was consistent with some studies in the field of GBL that learning support was seldom used by students (Bainbridge et al., 2022).

A potential approach to solving this issue is increasing the level of interactivity between students and the learning supports. For example, some researchers implemented modeling in the form of prompts with multiple choice questions (Adams & Clark, 2014; ter Vrugte et al., 2015), or explicit instructions with fill in the blank questions (Chen et al., 2012; Lee & Ke, 2019). In these instances, students are requested to answer questions or complete

tasks before proceeding to the next steps, thus actively engaging in studying these supports. It is believed that students' learning will increase when their engagement increases (Chi & Wylie, 2014).

Another technique to increase learning-support engagement is presenting *completion tasks*, which was proposed as an alternative of worked-out examples (van Merriënboer & Krammer, 1987). The difference between completion tasks and worked-out examples normally depends on the level of completeness (Wouters, 2017). Examples can be either complete or incomplete (Renkl & Atkinson, 2003). Complete ones (worked-out examples) provide students with all the procedural steps and solutions of the problem, while the incomplete ones (completion tasks) merely give specific steps or solutions for solving a particular part of the problem (Renkl, 2002; van Merriënboer & Krammer, 1987). From the cognitive load point of view, both forms of modeling have advantages and disadvantages. For example, when students possess prior knowledge, presenting a complete worked-out example might hamper students' learning processes. Because students need to analyze and integrate the redundant information with prior schemas in working memory, resulting in higher cognitive load (Kalyuga et al., 2003, 2012). On the other hand, compared to complete worked-out examples, it is likely that studying incomplete worked-out examples may help stimulate students' self-explanations (Renkl et al., 2004). As such, completion tasks can better engage students in active learning processes (van Merriënboer & Kirschner, 2018). Empirical studies supported the proposition that presenting completion tasks is superior to studying complete worked-out examples (e.g., Sweller et al., 2019; van Merriënboer & de Croock, 1992).

However, implementing modeling in the context of GBL seems more complex than in a conventional instruction. ter Vrugte et al. (2015) explored the role of modeling with distinct types of knowledge in the domain of math learning. The purpose of providing modeling in the game was to foster students' reflection towards specific aspects of domain knowledge. One hundred and forty-five students were randomly assigned to four conditions: the game with reflection prompts (completion tasks with conceptual knowledge only), the game with procedural prompts (completion tasks with procedural knowledge only), the game with reflection and procedural prompts (completion tasks with the combination of the conceptual and procedural knowledge), and the game without modeling (control group). The study findings revealed that GBL supported students' knowledge acquisition, but there was no significant effect of the types of learning supports on participants' math achievement. This study suggests that it is not only the form, but also the target type of knowledge that might mediate the level of cognitive processing elicited by modeling.

Prior research indicated that the use of modeling before the task trial could help learners construct a mental model (Barzilai & Blau, 2014) or serve as an advanced organizer for the directions of the subsequent learning tasks performance (Wouters, 2017), while modeling after the task trial aims to provide students with the opportunity of debriefing (Wouters, 2017), reflection (ter Vrugte et al., 2015), or self-explanation (Johnson & Mayer, 2010). However, the impact of the timing of modeling on learning remains a question. Barzilai and Blau (2014) conducted a pretest–posttest experimental design study to investigate the effects of modeling on students' performance of solving financial-mathematical word problems. The study compared a group without modeling, a group with modeling before the game started, and a group who received modeling just after playing the game. Their results revealed that the learners who received modeling before the game outperformed students in the other two conditions in the post-game math problem solving test.

In summary, despite the belief that modeling helps with schema construction (van Gerven et al., 2002), how to design and use modeling as a learning support in GBL

environments is still limited, inconsistent and inconclusive (Mayer, 2011; Wouters, 2017). Empirical research is warranted to examine different modeling approaches to better understanding *what*, *when*, and *how* to design such type of learning support in GBL environments in the domain of math learning.

Learning support and game flow

Game flow stems from the flow theory of Csikzentmihalyi (1975). It refers to a player's state of full immersion in the game playing. It is said to occur when a player is so involved in game play that the sense of space and time is lost (Squire, 2005). Game flow is a frequently used construct to describe game-play experience (Kiili et al., 2021; Procci et al., 2012) and indicate learners' immersion degree (Hou & Keng, 2021). Studies indicate that reaching game flow is important because it could positively impact learning performance (Chou et al., 2021), and is a primary factor on students' game enjoyment (Baumann et al., 2016; Sweetser & Wyeth, 2005).

On the other hand, game flow might inhibit players from being conscious or mindful about learning (ter Vrugte et al., 2015). It is reported or argued that students who are engaged in a GBL environment are likely to gain intuitive and implicit knowledge more than explicit knowledge (Johnson & Mayer, 2010; ter Vrugte et al., 2015). When students reach the state of game flow, they might be reluctant to actively perform cognitive learning processes and thus fail to integrate new knowledge learned in the game with their prior knowledge (ter Vrugte et al., 2015). As a result, prior studies suggest the importance of integrating learning support to transform intuitive processing into generative processing in GBL (ter Vrugte et al., 2015).

However, how to design external learning support that could foster generative learning without interrupting game flow and enjoyment remains misty. Some studies reported that integrating external scaffolds before the game did not interrupt or impact students' experience of flow (Barzilai & Blau, 2014). Other studies revealed that additional learning supports can decrease players' game flow, and may interrupt what is enjoyable about the game, which may have a negative effect on GBL (Erhel & Jamet, 2013; ter Vrugte et al., 2015; Vandercruysse et al., 2016). Consequently, the value of providing learning support will be optimal only if it facilitates the knowledge collection, organization and integration without interrupting the game flow (Ke, 2016; Vandercruysse et al., 2016). In other words, learning support needs to be intrinsic to the game mechanics to assist learning processes. This proposition is aligned with evidence-centered design in which students will opt to use learning support when it is nearly invisible to the game play (Mislevy et al., 2003). However, not all the deep content, strategies, and skills could be integrated to the game mechanics (Graesser et al., 2009). Consequently, such learning support should be carefully designed and incorporated in the GBL environment.

Research purpose

In this study, we investigated the effects of three types of game-based learning supports in the form of modeling (supporting conceptual knowledge only, procedural knowledge only, versus the combination of the two) for middle school students on their math achievement and perceived game flow. Two research questions were addressed:

- a) What are the effects of the type of learning supports (conceptual, procedural, versus conceptual and procedural) embedded in a learning game on students' math achievement?
- b) What are the effects of the type of learning supports (conceptual, procedural, versus conceptual and procedural) embedded in a learning game on students' perceived game flow?

Methodology

Research design

The study was conducted as a pretest–posttest experimental design with repeated measures, controlling the task complexity, to investigate the effects of the three different conditions of learning supports during a 4-day game-based learning workshop. The independent variable was the type of learning support (conceptual, procedural, and the combination of the two). Dependent measures included students' math achievement and perceived game flow.

Participants

Forty-five participants were recruited from four sixth-grade classes of a middle school in the western United States. Of those 45 students, 41 completed the entire game-based learning workshop. Specifically, 19 participants were male and 22 were female. Their ages ranged between 11 and 12 years old. The participants' demographic information is provided in Table 1.

Participants in each class were randomly assigned to one of the three conditions with different game-based learning supports: (a) learning support with conceptual knowledge only (LSCK), (b) learning support with procedural knowledge only (LSPK), and (c) learning support with both conceptual and procedural knowledge (LSCPK). A one-way ANOVA was conducted on the participants' pretest measurement, which we collected on Day One. The results showed that there were no significant differences in participants' prior knowledge of word problem solving in ratio and proportion knowledge among the three conditions ($F(2, 38) = 0.78, p > .05$).

Table 1 Students' demographic information

	LSCK group <i>N</i> = 13	LSPK group <i>N</i> = 14	LSCPK group <i>N</i> = 14
<i>Gender</i>			
Male	6 (46.2%)	6 (42.9%)	7 (50.0%)
Female	7 (53.8%)	8 (57.1%)	7 (50.0%)
<i>Ethnicity</i>			
African American	1 (7.7%)	0	1 (7.1%)
White American	5 (38.5%)	6 (42.9%)	6 (42.9%)
Hispanic	7 (53.8%)	8 (57.1%)	7 (50.0%)

LSCK: learning support with conceptual knowledge, LSPK: learning support with procedural knowledge, LSCPK: learning support with conceptual and procedural knowledge

Material

The research treatment included using two selected levels of an architecture game which consisted of the learning support activities and game-based math tasks. Specifically, when a player launched the game, he/she had to open the *Help* tab and select one of the three types of learning supports that was randomly assigned by the researcher before game play. After completing the activity of learning support, students proceeded to play the game-based math tasks to complete the game level. These materials are described below.

Game Used

ERebuild is a 3D, multi-level architecture game that aims to promote math problem solving and learning for middle school students, with a focus on the math topics of ratios and proportional reasoning, geometric figures (de)composition and properties calculation, expressions and equations (Kim et al., 2021). The overall goal of the game is to reconstruct a neighborhood after a natural disaster. The latest version of the game involves multiple episodes (i.e., *Island*, *Desert*, *School*, and *Farm*) that convey a variety of game-based math tasks. Besides the progressive difficulty of the math topics, the major difference among the four episodes is the simulated contexts or scenarios of problem solving. In general, the participants are requested to complete each level of the design quest that simulates an authentic problem situation through the application of math knowledge. For example, the participants need to buy materials (e.g., blocks) to construct a specific building to shelter all surviving families after a disaster damage in the episode of *Desert* (see Fig. 1).

In the current study, two levels of challenging quests in the *School* episode were selected as the major game-based math tasks. These two levels of quests were chosen because: (a) the math content embedded focused on the topics of ratio reasoning and area; (b) prior user tests indicated that students had difficulty solving these math-based game tasks. In the selected two game levels, students need to trade the items in the store, by gauging the



Fig. 1 Screenshot of a game level in ERebuild

residents' needs for the living space and then the amount and cost of the materials needed (see Fig. 2a). They were then required to construct the designated building through the acts of object placing, moving, and rotating (see Fig. 2b).

Game-based math task in the selected level

The math tasks examined in the selected two game levels are in line with the Common Core State Standards (CCSS) of Mathematics for middle school students on ratio reasoning: “understand ratio concepts and use ratio reasoning to solve problems” (Common Core State Standards Initiative, 2010, p. 41) and on geometry: “solve real-world and mathematical problems involving area, surface area, and volume” (Common Core State Standards Initiative, 2010, p. 44). The difference between the two game-based math tasks is whether unit rates are associated with integers or ratios of fractions. For example, in the first game-based math task, numbers are given in the form of whole numbers (e.g., a math student requires 3 m²). In the second game-based math task, the ratio number is given in the form of a fraction (e.g., the ratio of computer and writing student need is 1:1/2), which requires students to use the knowledge of fraction of ratio and proportional reasoning to complete the game-based math problem solving task.

To successfully solve the target game-based math tasks in the selected levels, the player must (1) measure the length and width of the container to calculate the area that a container has, (2) calculate the area that each type of residents (i.e., math students, literacy students, and computer science students) needs via ratio reasoning, and (3) calculate how much area that all residents need.

Learning support design in the selected level

Three types of learning supports were developed. Conceptual knowledge involves an understanding of the interrelations of each relevant information element (e.g., Byrnes, 1992; Hallett et al., 2012). Therefore, the purpose of learning support with conceptual knowledge only (LSCK) is to facilitate the learner to construct a mental representation by identifying relevant and useful information about the problem. Prior studies indicated that qualitative relationship mapping (e.g., semantic network and concept map) is a valid approach to developing students' mental models (Byrnes & Wasik, 1991; Quintero, 1983; Sarwar & Trumpower, 2015). Correspondingly, players in the current study were prompted to drag and drop task-relevant information (e.g., ‘How many containers are needed’; ‘Area

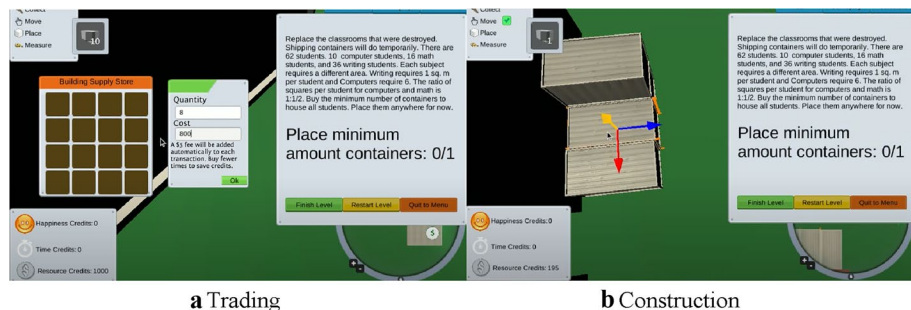


Fig. 2 Game mechanism of trading and construction in the selected game level

all students need') to make an appropriate relationship chart (see Fig. 3). Players would not proceed to the next step until they successfully completed the relationship chart. In the next step, students got to solve multiple-choice questions that prompt them to select a mathematical model or expression for each mapped relationship. These questions directed players' attentions to the most important parameters and their relations when solving the game-based math task. Upon trying every multiple-choice question, players would receive summative feedback. Figure 4 shows examples of such prompts.

In comparison, procedural knowledge is conceptualized as a "goal-oriented action sequence" (Byrnes, 1992, p. 236). As a result, learning support with procedural knowledge only (LSPK) is aimed to help learners generate and execute a computational solution. It offers information on the computational steps for solving game-based math tasks. It is constituted by both step-by-step procedures of game-based math problem solving and the symbolic representations used in the procedures. Figure 5 illustrates examples of different computational steps for a game-based math task.

The aforementioned LSCK and LSPK are combined for the learning support with both conceptual and procedural knowledge (LSCPK) group (i.e., students complete tasks of LSCK first and then proceed to tasks of LSPK). Except for the difference in the types of learning supports, there is no other difference in the game-based math learning activities across the three study groups.

Based on the findings from prior research in the literature, learning support in the form of modeling in the current study had the following design characteristics: (a) featuring

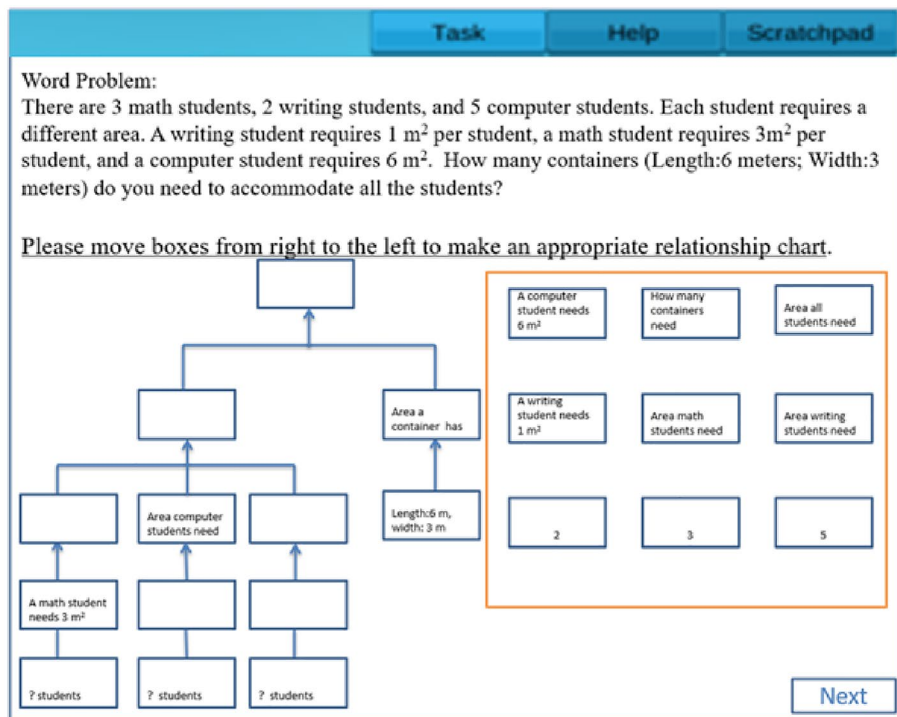


Fig. 3 Screenshot of the task of the relationship chart (part of LSCK)

Task **Help** **Scratchpad**

Word Problem:
 There are 3 math students, 2 writing students, and 5 computer students. Each student requires a different area. A writing student requires 1 m^2 per student, a math student requires 3 m^2 per student, and a computer student requires 6 m^2 . How many containers (Length:6 meters; Width:3 meters) do you need to accommodate all the students?

Great job! Based on relationship chart you built, let's plan what solution you will use.

Choose the best answer for Step **A**

What solution is used to get how many containers are needed?

a. Addition
 b. Subtraction
 c. Multiplication
 d. Division

Previous **Next**

Task **Help** **Scratchpad**

Word Problem:
 There are 3 math students, 2 writing students, and 5 computer students. Each student requires a different area. A writing student requires 1 m^2 per student, a math student requires 3 m^2 per student, and a computer student requires 6 m^2 . How many containers (Length:6 meters; Width:3 meters) do you need to accommodate all the students?

Great job! Based on relationship chart you built, let's plan what solution you will use.

Choose the best answer for Step **a**

What solution is used to get how many areas that all students need?

a. Addition
 b. Subtraction
 c. Multiplication
 d. Division

Previous **Next**

Fig. 4 Examples of multiple-choice questions (part of LSCK)

explicit instruction with high interactivity (i.e., relationship chart completion, multiple choice questions, and fill in the blank questions), (b) being given before the game-based task, and (c) being completion tasks.

Research procedures

On Day One, a 15-minute pretest was administrated with all participants to check if there was difference with respect to students' prior knowledge in solving math problems corresponding to ratios and proportional reasoning. On Day Two, a 15-minute tutorial was presented to the participants on how to play the game with the mouse and the keyboard, as

TaskHelpScratchpad

Word Problem:
 There are 3 math students, 2 writing students, and 5 computer students. Each student requires a different area. A writing student requires 1 m² per student, a math student requires 3m² per student, and a computer student requires 6 m². How many containers (Length:6 meters; Width:3 meters) do you need to accommodate all the students?

Please fill in the boxes below to get the correct answer.

Step I: Compute area math students need

1	X 3	3
3	=	?
	X	

So all computer students need m²

PreviousNext

TaskHelpScratchpad

Word Problem:
 There are 3 math students, 2 writing students, and 5 computer students. Each student requires a different area. A writing student requires 1 m² per student, a math student requires 3m² per student, and a computer student requires 6 m². How many containers (Length:6 meters; Width:3 meters) do you need to accommodate all the students?

Please fill in the boxes below to get the correct answer.

Step b: Compute area a container has

	X		=	area of a container
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Length X Width = Area

So the area of a container is m²

PreviousNext

Fig. 5 Examples of fill in the blank questions (part of LSPK)

well as the overall goal of the game. All participants then played the training levels of the game for about 45 minutes, aiming to provide participants with opportunities for practicing operating skills and getting familiar with the game actions and objectives. On Day Three and Day Four, individual participants were randomly assigned to one of the three study groups. Participants in each group spent 45 minutes playing the intervention (i.e., 20-minute learning support activity and 25-minute game-based math task, see Table 2), followed by a 15-minute posttest. Specifically, they completed the first game-based math task with Posttest 1 on Day Three, and the second game-based math task with Posttest 2 on Day

Table 2 Intervention design of the study

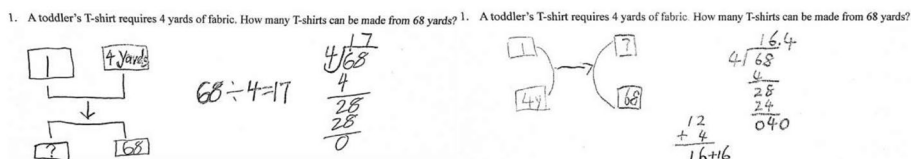
Activity	Sub-task (#)	LSCK	LSPK	LSCPK	Time
Learning support activity #1	Relationship chart (1)	X		X	20 minutes
	Multiple-choice question (6)	X		X	
	Fill in the blank question (6)		X	X	
Game-based math task # 1		X	X	X	25 minutes
Learning support activity #2	Relationship chart (1)	X		X	20 minutes
	Multiple-choice question (6)	X		X	
	Fill in the blank question (6)		X	X	
Game-based math task # 2		X	X	X	25 minutes

LSCK: learning support with conceptual knowledge, LSPK: learning support with procedural knowledge, LSCPK: learning support with conceptual and procedural knowledge

Four. On Day Four, all the students also completed a game flow questionnaire after game play and before Posttest 2.

Instrument

In this study, math achievement was measured by participants' performance with three isomorphic sets of test items (i.e., one pretest and two posttests). Each test contained three math word problems that were developed based on CCSS of Mathematics for sixth graders on ratio reasoning and geometry. To increase the reliability of the tests, two in-service math teachers at the selected middle school were consulted to confirm the quality of the three sets of the test items. A rubric was used to grade each participant's knowledge acquisition. Specifically, students were required to write down a conceptual relationship chart and procedures for each step as detail as possible. Each test item was decomposed into different step-by-step procedures. For each step, students were scored 1 point for correctly providing the answer, 0.5 points for giving the partially correct answer, and 0 point for giving the wrong answer or no information. Figure 6 illustrates two examples of fully correct answer and partially correct answer. The first two test items were worth 3 points in total, respectively and the third test item was 10 points due to distinct and complex procedures after decomposing. Overall, the total scores of each test set were ranged from 0 to 16. The internal consistency reliabilities for the three sets of tests (i.e., Pretest, Posttest 1, and Posttest 2) were 0.57, 0.69, and 0.69 respectively. These low internal consistency reliabilities were mainly due to the small number of test items.

**Fig. 6** Examples with full scores (left side) and partial scores (right side)

A seven-item, five-point Likert scale questionnaire was used to measure participants' perceived game flow. To avoid interrupting the game-play flow, this questionnaire was delivered right after game play instead of during game play. The items selected were derived from the GameFlow questionnaire developed by Kiili (2005). The original flow scale consisted of a total of 23 items, aiming to evaluate nine dimensions of players' game flow experience: challenge, goals, feedback, playability, control, rewarding experience, concentration, self-consciousness and time distortion (Kiili, 2005). Since the target audience of this study were middle school students, two dimensions of the instrument (feedback and playability) were removed because most of the children cannot fully understand the relevant questions (Inal & Cagiltay, 2007). This customized questionnaire had high reliability in previous studies, ranging from 0.78 to 0.83 (e.g., Inal & Cagiltay, 2007; Kiili, et al., 2014), and the reliability for the current study was 0.78, which indicated a satisfactory level of internal consistency.

Data analyses

A mixed-design ANOVA with repeated measures was used to evaluate differences among the repeated measurements of math tests in the three groups. The three learning support conditions were used as the between-subject factor, and the repeated measurements of participants' math performance were the within-subject factor. The pretest–posttest scores of the three groups were compared to determine the effectiveness of the types of learning supports on students' performance of game-based math problem solving.

A one-way ANOVA was used to measure if the treatment effect of learning support was present on students' game flow. The result of participants' questionnaire was used as the dependent variable and the group condition was the independent variable for this analysis.

Results

Effects of learning supports on math achievement

The result of the mixed-design ANOVA with repeated measures indicated that there was a significant interaction between the time of measurement and the type of learning support in affecting students' math achievement, $F(4, 76) = 4.04$, $p < .05$, $\eta^2 = 0.18$. The result indicated that students' math test scores among the three groups differed significantly in changes from Pretest to Posttest 1 and to Posttest 2, $F(2, 76) = 46.78$, $p < .05$, $\eta^2 = 0.55$. The descriptive statistics of each group's math performance in the three tests is demonstrated in Table 3. Further analysis—an ANOVA with simple effects—was conducted to analyze within-group differences separately. The results of pairwise comparisons using the Bonferroni method on the math performance of the study groups across the three time points indicated that students in all three groups had significantly increased math test mean scores from Pretest to Posttest 1 (i.e., LSCK group $p = .02$; LSPK group $p < .05$; LSCPK group $p < .05$) and from Pretest to Posttest 2 (i.e., LSCK group $p = .02$; LSPK group $p < .05$; LSCPK group $p < .05$), though there were no significant differences from Posttest 1 to Posttest 2 (i.e., LSCK group $p = 1.00$; LSPK group $p = 0.32$; LSCPK group $p = 0.32$) across the three groups. Figure 7 illustrates the differences in math performance among the three learning-support groups across the three times of measurement.

Table 3 Mean results and standard deviations for three times of measurements of math test in the three conditions

Test	LCK <i>N</i> = 13	LSPK <i>N</i> = 14	LSCP <i>N</i> = 14
	Mean (SD)	Mean (SD)	Mean (SD)
Pretest	3.46 (2.02)	4.64 (1.95)	4.21 (1.95)
Posttest 1	5.54 (3.52)	10.29 (3.40)	7.50 (3.40)
Posttest 2	5.54 (3.18)	11.00 (3.07)	8.21 (3.07)

LCK: learning support with conceptual knowledge, LSPK: learning support with procedural knowledge, LSCP: learning support with conceptual and procedural knowledge

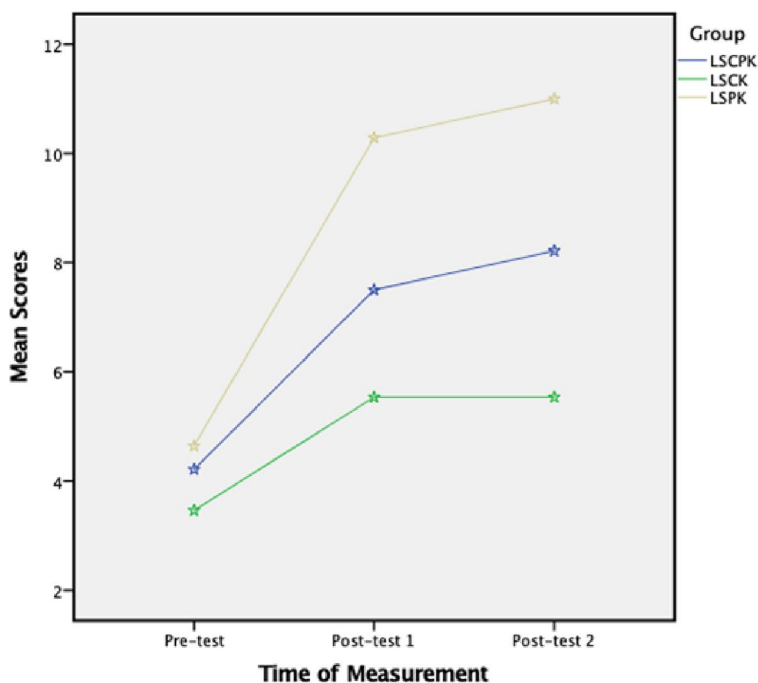


Fig. 7 Math test performance among types of learning supports across time of measurements

The result of post hoc comparison of three experimental groups with respect to students' math test scores indicated that LSPK was significantly more effective than LSCP, and LSCP was significantly more effective than LCK, in promoting the overall math test performance. In other words, LSPK was the most effective in promoting students' overall math test scores, followed by LSCP and then LCK.

Effects of learning supports on game flow

The result of one-way ANOVA analysis showed that there was no significant difference among the three groups on their perceived game flow, $F(2.38)=0.56$, $p=0.58$. The game flow scale was numerically coded with 1–5, representing “Disagree strongly” to “Agree strongly”. Table 4 presents students’ self-reported game flow results, in which the mean scores fall between 3.00 and 4.43 across the three groups.

Discussion

The present study adopted a value-added approach in game-based learning to investigating how diverse types of learning supports in the form of modeling, could influence students’ math performance and game flow experiences. The results indicated that all students improved in their math performance as time passed by and had positive dispositions toward their game flow experience during game play. However, students who participated in LSPK condition demonstrated a significantly higher level of math performance consistently over time than those who were provided LSCPK and LSCK.

Students’ math performance

The results showed that, after playing the game with diverse types of learning supports, students’ overall math performance was significantly higher than that before the game. The study findings confirmed that if students are constantly engaged in studying learning support in the form of modeling for math problem solving in GBL, their math performance would increase over time. This is in line with the report of prior research on the positive effects of GBL with learning support on math learning (e.g., Adams & Clark, 2014; Lee & Ke, 2019; Vandercruysse et al., 2016).

On the other hand, this finding contrasts with the work by ter Vrugte et al. (2015) who reported that modeling with conceptual knowledge and/or with procedural knowledge failed to improve students’ math achievement. It should be noted that in the study by ter Vrugte et al. modeling was provided after students’ first failed attempt and worked as self-explanation (Johnson & Mayer, 2010), whereas in the current study, modeling

Table 4 Mean scores and deviations of the game flow experience dimensions

Dimension	LSCK		LSPK		LSCPK	
	<i>N</i> = 13		<i>N</i> = 14		<i>N</i> = 14	
	Mean	SD	Mean	SD	Mean	SD
Overall flow	27.62	3.97	28.86	3.23	27.64	3.37
Challenge	3.77	0.73	3.57	0.65	3.50	0.76
Goals	3.69	0.86	3.00	1.11	3.71	0.61
Concentration	4.31	1.32	4.43	0.85	4.07	1.21
Time-distortion	3.38	1.45	4.07	1.07	4.00	1.04
Self-consciousness	3.69	0.86	4.00	0.88	4.14	0.66
Rewarding experience	3.23	1.24	3.71	1.07	3.57	0.94
Control	3.54	0.78	3.07	0.62	3.64	0.63

was presented before students tackle the tasks. In other words, the provision of modeling before game-based tasks may have helped students construct a mental model that guided the direction of the subsequent learning actions (Barzilai & Blau, 2014; Wouters, 2017). Different timing of modeling provision serves distinct functions, which may explain the difference in the findings. Another potential account for the contradictory result is related to students' active or passive model processing (Renkl, 2002; Renkl & Atkinson, 2003). Although incomplete examples were provided in ter Vrugte et al.'s study, students only needed to click on the 'OK' button when they finished reviewing the examples provided. It is possible that in ter Vrugte et al.'s study students were involved in passive model processing. In comparison, students of the current study were required to actively engage in studying incomplete examples by completing a relationship chart, answering multiple choice questions, and/or filling in the blank questions. As such, the current finding supports the proposition that increasing students' processing of the support will promote learning (Chi & Wylie, 2014).

The study results indicated that diverse types of game-based learning supports had different effects on students' math learning outcome. This finding is consistent with the previous research finding that the design of learning support in GBL mediated the development of quantitative proficiency and qualitative understanding (Lee & Ke, 2019). In the current study, students in the condition of LSPK constantly showed significantly higher post-test scores after game play than the other two conditions, whereas LSCK was constantly least effective in improving students' math achievement. There could be distinct reasons to account for this result. First, the design of LSPK seemed more effective in fostering students' procedural knowledge acquisition for mathematical problem solving, but the design of LSCK (i.e., the conceptual relationship chart) might fail to promote students' conceptual knowledge development for problem solving. Consequently, students in the LSCK and LSCPK conditions, compared with students in the LSPK condition, could experience higher cognitive load when processing information of the relationship chart that could be abstract or novel to them (Sweller et al., 1998). Second, treatment effects could be a potential explanation of why LSCK condition was least effective to improve students' math test scores (Barzilai & Blau, 2014). In this study, students were only exposed twice in short periods (i.e., 20 minutes per game level) to each learning support. Students might grasp the procedural knowledge faster than the conceptual knowledge within a limited time (Hallett et al., 2012; Usta & Yilmaz, 2020). Therefore, we recommend that researchers extend studies by examining if there is any relation between students' in-game performance measured by on-task time and time of attempts and the speed of knowledge proficiency. It is also possible that students need extra-training to transfer the learned conceptual knowledge from the game-based math tasks to the conventional math assessment. Lastly, the three sets of math tests may not fully capture the conceptual knowledge acquired from the intervention, even though the tests are expected to measure an integrative application of both conceptual and procedural knowledge.

Students' game flow

The game-flow survey results suggested that students in each group had positive dispositions toward their game flow experience during game play. The findings supported Barzilai and Blau (2014) who reported that providing learning support would not interrupt game flow experience. Although there were no statistical differences among the three learning support conditions in terms of promoting positive students' game flow, we noted that there

were numerical differences that could point the way toward future research. For example, descriptively students in the LSPK group had the highest overall game flow experience, followed by LSCPK and LSCK group, which could indicate that reaching game flow may positively impact learning performance (Chou et al., 2021). This finding is aligned with the finding governing the impact of learning support conditions on the math test performance. Meanwhile, students in the condition of LSPK had difficulty in understanding what they need to do to achieve the game objectives. Their perceived experience of game control was also lower than that of their peers. Thus, further study is needed to investigate what factors will cause differences in the dimensions of 'goal understanding' and 'game control'.

In summary, it can be argued that integrating learning support before game-based math tasks did not affect students' experience of game flow, but the learning support should be carefully designed and incorporated in the learning environment. Specifically, we recommend that researchers design learning supports that are intrinsic to the game through evidence-centered design (Mislevy et al., 2003) or similar integrated game assessment design approaches.

Implications

Based on the study findings, we propose the following design heuristics governing game-based learning supports in the form of modeling for GBL. First, there is strong evidence that students' math performance can substantially differ by integrating distinct types of learning supports in GBL. The current study findings indicate that participants in the condition of LSPK have achieved better than the students in the condition of LSCPK, which contradicts the universal acceptance of "the more the better". Second, the current study suggests that content-specific learning support in the form of modeling should be given before the game-based learning task to avoid the interruption of game flow or cognitive overload during game play. In addition, to avoid passive and superficial learning, students should be given more opportunities to process the supports, such as increasing the level of interactivity between students and the learning material, as well as extending the studying period and frequency.

Limitations and future research

The study findings provide an initial insight into the potential benefits of diverse types of learning supports embedded in the GBL environment for students' math performance and perceived game flow. Although this study has demonstrated that adding learning support in the form of modeling could positively influence students' math performance and game flow experiences in the context of GBL, the study has its limitations. First, the current study has not involved a large sample or a control group. Hence, the study should be replicated with a larger sample size and employ a control-group design to further explore if learning support embedded in the GBL, compared with the GBL without learning support, promotes knowledge acquisition of math problem solving. Second, we should highlight that this study adopts a value-added approach to GBL so the design of learning support in the form of modeling is not intrinsically embedded in the game world. Future research may place more emphasis on design and develop intrinsic-embedded learning support in GBL. It can be achieved by incorporating evidence-centered design by integrating learning mechanics

and learning actions into game mechanics so that learning will become an intrinsic part of gameplay. Third, we recommend the use of unobtrusive method of capturing participants' game flow in situ instead of post-game reflection in future studies. For example, Flowometer (Reese, 2016) could be a powerful technique used to collect a time-based measure of players self-perceptions of gameplay experience. It is also warranted to create contextualized assessments to delimitate the phenomenon of context-dependent memory under distinct conditions. Furthermore, we strongly recommend that researchers should collect and analyze students' in-game performance to provide a detailed and deep understanding of how they perform in the process of game-based problem solving. Future research could examine if computational fluency, as a prerequisite for students' knowledge acquisition of math problem solving, may mediate their game-based learning with diverse types of learning supports. In general, future research should further investigate the design of effective learning support that will balance the acquisition of domain-specific knowledge and game flow.

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Data availability The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This study was conducted with the university IRB (human subject protection) approval and adhered to ethical guidelines as the nature of study demanded. Being constrained by the human subject protection policies, the original study data are not open.

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